

Shen (Sam) Xie

Email: samxiexs@gmail.com sxie322@gatech.edu
GitHub: [samxiexs](https://github.com/samxiexs) LinkedIn: [shenxie](https://www.linkedin.com/in/shenxie) Website: samxiexs.github.io

EDUCATION

- **Georgia Institute of Technology, Atlanta, United States** (Aug 2026 – Present).
Master's, Major in Computational Science and Engineering; H. Milton Stewart School of Industrial and Systems Engineering.
- **Wuhan University, Wuhan, China** (Sep 2020 – Jun 2024).
Bachelor's, Major in Politics, Philosophy, and Economics; Minor in Computer Science and Engineering; Hongyi Honor College.
Honor Graduate; First Prize Academic Scholarship (ranked 2nd of 53).

RESEARCH INTERESTS

- § AI for Science; § Multimodal Machine Learning; § Human-Centered AI; § Computational Social Science.

RESEARCH EXPERIENCE

- **HEC Paris, Paris, France** (Mar 2025 – Aug 2026).
Remote Research Assistant, Department of ISOM; supervisor: Prof. Xitong Li.
 - Supported research on **machine learning and human-AI interaction** in digital platforms.
 - Built real-time data pipelines for **Reddit, SSRN, FastMoss, TikTok Shop**, producing research datasets for creator-economy, platform consumer behavior, and digital marketing studies.
 - Designed influencer-product matching workflows with rule-based and machine-learning approaches.
- **Hong Kong University of Science and Technology, Hong Kong SAR, China** (Aug 2024 – Jul 2025).
Onsite Research Assistant, Department of ISOM; supervisor: Prof. Tatko Koh.
 - Supported theory development and empirical design through literature synthesis on **crowdsourcing contests, human-AI interaction, and platform strategy**; constructed a large-scale **99designs** dataset for empirical analysis.
- **Wuhan University, Wuhan, China** (Feb 2024 – Jun 2024).
Research Assistant, Department of MSE; supervisor: Prof. Baojun Gao.
 - Conducted robustness checks and **Structural Topic Modeling** for a **Journal of Retailing (ABS 4)** paper in its second round of review using TripAdvisor review data.

SELECTED RESEARCH PROJECTS

- **Speech-Text Multimodal Modeling for Psychosis and Suicide-Risk Assessment** (Sep 2026 – Present).
In progress.
 - Developing **speech-text multimodal models** for psychiatric assessment using clinical interviews with people experiencing psychosis and crisis-line calls involving suicide risk.
 - Combining transcripts and audio to model **linguistic and acoustic cues**, with data privacy and careful evaluation as priorities; the goal is to support clinicians and crisis responders.
- **LLM Agents for Scientific Experiment Selection** (Sep 2026 – Present).
Co-first author with Yuming Zeng; Work in progress; planned submission to ICML 2027.
 - Developing a framework for **adaptive scientific experiment selection**, where LLM agents use experimental histories to select informative experiments under limited experimental budgets.
- **EEG-to-Speech Decoding via Cross-Modal Representation Learning** (Apr 2026 – Present).
First author with Mingwei Sun, Shijie Zhang, and Xitong Li; Work in progress; planned submission to ICML 2027.
 - Developing an **EEG-to-speech-content** framework that maps variable-channel EEG signals to speech representations using coordinate-aware CNN-Conformer encoders.
 - Building leakage-controlled evaluation across subjects, speech content, and datasets, with counterfactual EEG controls to assess whether decoding reflects neural signals rather than memorization.
- **Diffusion-Based Image-Audio Alignment** (Aug 2026 – Present).
Third author with Biyang Xin, Mingwei Sun, Shijie Zhang, and Xitong Li; In progress.
 - Contributing to a diffusion-based framework for learning aligned **image-audio representations** for cross-modal correspondence and generation.
- **Vision-Language Models for Product Advertising** (Nov 2025 – Aug 2026).
Second author; under review at CHI 2027.
 - Manuscript title and further details withheld during peer review.

SELECTED RESEARCH PROJECTS (CONTINUED)

- **Content Similarity in Recent Reviews and Hotel Revenue** (May 2025 – Present).
Second author with Baojun Gao and Jiawen Wang; Manuscript in preparation; planned submission to International Journal of Hospitality Management (ABS 3); [GitHub](#).
 - Analyzes **1.5M TripAdvisor reviews** and revenue data from **2,990 hotels** across Texas cities.
 - Preliminary analyses suggest a negative association between adjacent review similarity and hotel revenue, with variation by hotel segment, review volume, and rating.

SELECTED RESEARCH & ENGINEERING PROJECTS

- **Influencer Agent for TikTok Shop** (Sep 2025 – Aug 2026).
with InFound.ai; [GitHub](#).
 - Developed backend services and experimentation tools for an **influencer marketing system** on macOS and Windows, supporting creator search and profiling, automated outreach, sample management, and sales-data processing.
 - Built data-processing services connecting creator matching, recommendations, and operational analytics through API and MySQL integration.
 - Received the **Gold Award** at the Guangdong-Hong Kong-Macao Greater Bay Area Inno Awards Demo Day.
- **Ad Recommendation with RQ-VAE Semantic IDs and RankMixer Bidding** (Oct 2025 – Mar 2026).
with an advertising company in Shenzhen, China.
 - Trained an **RQ-VAE** to compress joint image-text embeddings into **semantic IDs** using four 256-entry codebooks, with **over 92%** utilization in each codebook. [GitHub](#)
 - Implemented a **RankMixer** bidding reranker with joint click-through and conversion prediction; offline evaluation showed an **AUC improvement of 0.5%**. [GitHub](#)
- **Social Science Paper Crawler and Analysis Agent** (Apr 2025 – Jun 2025).
with Ziqing Li; [GitHub](#).
 - Built a crawling agent and dataset covering more than **1.2 million papers**, including titles, abstracts, author information, and citation links.
 - Designed the pipeline to support downstream topic discovery, citation analysis, and automated literature exploration.

ACADEMIC ACTIVITIES

- Chinese Open Science Network Hackathon 2025 (Aug 2025). *Invited by Southern University of Science and Technology and University of Macau.*
- Quant Marketing PhD Summer Workshop (Jul 2025). *Invited by City University of Hong Kong.*
- AI for Business Summer Workshop (Jul 2025). *Invited by University of Science and Technology of China; **Outstanding Participant Award**.*
- Paper Sharing: Unexpected Negative Effect from Platform Regulation (May 2025). *Management Science Seminar co-hosted by Wuhan University and Xi'an Jiaotong University.*

SKILLS

- Programming and data tools: **Python, R, Stata, SQL, $\text{\LaTeX}$** .
- Engineering stack: **FastAPI, Redis, RabbitMQ, Docker, PostgreSQL, MySQL**.
- Methods: multimodal learning, recommender systems, NLP, structural topic modeling, and large-scale web data collection.